AGRIKRANTI 1928

Aditya Kumar, M.Sc. (Ag.) Soil Science - Gold Medalist (DDUGU) · Semester Paper Series

B.Sc. (Hons.) Agriculture - First Semester Examination

MAT-101: Elementary Mathematics · Credit 2 (1+1)

Model Question Paper - 1 (Previous Year Pattern, syllabus based)

Time: 2 Hours **Maximum Marks: 70**

Note: (i) Attempt any **five** questions from Section A and any **three** questions from Section B. (ii) Answer each question of Section A within 50 words. खंड A से कोई पाँच तथा खंड B से कोई तीन प्रश्न कीजिए।

Section A (Attempt any five) 5 × 5 = 25

1. Find the coordinates of the point which divides the join of (2, 3) and (5, -3) externally in the ratio 2 : 1. **5**

2. Find the angle between the lines $y = \sqrt{3}\,x + 5$ and $\sqrt{3}\,y = x - 6$. **5**

3. Find the centre and radius of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$. **5**

4. Find the derivative of $\sin x$ from first principle. **5**

5. Differentiate x^x with respect to x. **5**

6. Evaluate: $\lim_{x \to 2} \dfrac{x^3 - 8}{x - 2}$ **5**

7. Evaluate: $\int x\, e^x\, dx$ **5**

8. If $A = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$, show that $A^2 - 5A = 2I$. **5**

Section B (Attempt any three) 3 × 15 = 45

1. (i) Find the equation of the circle passing through (0, 0), (2, 0) and (0, 4). (ii) Find the equations of the bisectors of the angles between $3x + 4y - 7 = 0$ and $12x - 5y + 8 = 0$. (iii) Show that $y = x + 5\sqrt{2}$ touches the circle $x^2 + y^2 = 25$. **15**

2. Differentiate w.r.t. x: (i) $\dfrac{x^2 + 1}{x - 1}$ (ii) $\tan^{-1}\left(\dfrac{2x}{1 - x^2}\right)$ (iii) $(\sin x)^{\cos x}$ **15**

3. (i) A farmer has 400 m of fencing to enclose a rectangular field. Find the dimensions that give the maximum area. (ii) Find the maximum and minimum values of $y = x^3 - 6x^2 + 9x + 1$. **15**

4. Evaluate: (i) $\int x^2 \log x\, dx$ (ii) $\int \dfrac{2x}{1 + x^2}\, dx$ (iii) $\int_0^{\pi/2} \sin^2 x\, dx$ (iv) Find the area under $y = x^2$ from $x = 0$ to $x = 3$. **15**

5. (i) Find the inverse of $A = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{vmatrix}$ (ii) Prove that $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a - b)(b - c)(c - a)$ **15**